

EDA Report: Customer Churn Prediction and Analysis

Milestone 1: Data Collection, Exploration, and Preprocessing

Objective

To explore and clean a customer churn dataset to ensure data quality and readiness for predictive modeling. This includes identifying missing values, anomalies, inconsistencies, and relevant patterns that can guide feature engineering and model development.

1. Data Collection

- The dataset was loaded from a public GitHub repository.
 - It contains a mix of customer demographic data, behavioral metrics, subscription attributes, and a target column: `churn_risk_score`.
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2. Initial Exploration

- Sample inspection revealed unusual values:
 - `'xxxxxxxx'` in the `referral_id` column.
 - `'Error'` values in `avg_frequency_login_days`, which should be numeric.
- Dataset Information:
 - Presence of object-type columns containing numeric errors.
 - Key columns with missing values: `region_category`, `preferred_offer_types`, `points_in_wallet`.
- Statistical summary indicated:
 - Potential outliers.

- Negative values in features where only positive values make sense (e.g., `avg_time_spent`).
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3. Data Cleaning and Preprocessing

- **Handling Invalid & Missing Values:**

- Replaced invalid entries like `'Error'` and `'?'` with `NaN`.
- Converted `avg_frequency_login_days` to float and replaced negatives with `NaN`.
- Removed rows with missing values in `churn_risk_score` (target column).

- **Dropped Irrelevant Columns:**

- `customer_id`, `Name`, `security_no`, `referral_id`.

- **Date Features:**

- Parsed and sorted date columns: `joining_date` and `last_visit_time`.

- **Negative Value Handling:**

- Replaced negative values in `avg_time_spent`, `points_in_wallet`, and `days_since_last_login` with `NaN`.
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4. Distribution Analysis

- **Histograms & KDE Plots:**

- Many numeric columns showed skewed distributions and outliers.

- **Boxplots & Violinplots:**

- Confirmed the presence of outliers.
- Applied log transformation to `avg_time_spent` to reduce right skewness.

- **Imbalance in Target:**

- `churn_risk_score` shows class imbalance, which needs to be considered during modeling.
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5. Missing Value Analysis

- Visualized missing data using:
 - Bar plot (with both counts and percentages).
 - `missingno` matrix and heatmap.
 - No strong evidence of correlated missingness between columns.
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6. Key Insights

Aspect	Observation
Data Quality	Some columns contained string-based errors; handled via cleaning.
Target Column	<code>churn_risk_score</code> had invalid (-1) values removed.
Outliers	Present but retained to preserve potentially meaningful variance.
Feature Engineering	Log-transformed skewed features and removed irrelevant features.
Missingness	No systemic missingness patterns; imputation can proceed independently.
Class Imbalance	Detected in <code>churn_risk_score</code> ; modeling will require balancing techniques.

Sanity Check: referral_id vs joined_through_referral

A consistency check was conducted to validate the relationship between the `referral_id` and the `joined_through_referral` fields. Logically:

- If a user was marked as "joined through referral," then a referral ID should exist.
- If a user was not referred, the referral ID should be missing or clearly invalid.

However, inconsistencies were found. Some users marked as referred had the placeholder `"xxxxxxxx"` as their referral ID. These anomalies raise concerns about data quality.

Rather than removing these rows outright, the recommendation is to consult the data owners to clarify what `"xxxxxxxx"` represents. It may be a placeholder or an anonymized value. Until then, these rows should be handled carefully, possibly by treating `"xxxxxxxx"` as a separate category in downstream tasks.

Next Steps

- Impute missing values (mean/median/mode or model-based).
- Encode categorical variables (e.g., `gender` , `region_category` , `preferred_offer_types`).
- Engineer new features and perform correlation analysis.
- Begin initial model development and evaluation.